

Clinic Management System - Test Cases

Module 1: Patient Information Management

ID	Module	Function / Method	Description	Input / Condition	Expected Result
TC01	Patient	Clinic.registerPatient	Register a valid patient	Valid Patient object	Patient added to patients list & JSON saved (Approved)
TC02	Patient	Receptionist.registerPatient	Receptionist creates patient	Valid inputs (ID, name, etc.)	Returns new Patient object (Approved)
TC03	Patient	Patient.setFullName	Validate name regex	Input: "John123" (Numbers)	Throws IllegalArgumentException (Approved)
TC04	Patient	Patient.setPhone	Validate phone length	Input: "099" (3 digits)	Throws IllegalArgumentException (Must be 10 digits) (Approved)
TC05	Patient	Patient.setAddress	Validate empty address	Input: "" (Empty string)	Throws IllegalArgumentException (Approved)
TC06	Patient	Receptionist.updatePatientInfo	Update contact info	Valid Patient, new Phone/Addr	setPhone & setAddress called; savePatientChanges called (Approved)

TC07	Patient	Clinic.searchPatient	Search existing patient	Valid patientId	Returns correct Patient object (Approved)
TC08	Patient	Clinic.searchPatient	Search non-existent patient	Invalid patientId	Returns null (Approved)
TC09	Patient	Patient.addMedicalRecord	Add history record	Valid MedicalHistory object	Record added to medicalHistory list (Approved)
TC10	Patient	Patient.viewMedicalHistory	View empty history	Patient with no history	Prints "No hay registros medicos disponibles" (Approved)

Module 2: Doctor Information Management

ID	Module	Function / Method	Description	Input / Condition	Expected Result
TC11	Doctor	Clinic.addDoctor	Add new doctor	Valid Doctor object	Doctor added to doctors list & JSON saved (Approved)

TC12	Doctor	Clinic.searchDoctor	Search existing doctor	Valid doctorId	Returns correct Doctor object (Approved)
TC13	Doctor	Clinic.searchDoctor	Search invalid doctor	Invalid doctorId	Returns null (Approved)
TC14	Doctor	Doctor.viewAppointments	Filter appointments by Doctor	Clinic object with appointments	Prints only appointments where app.getDoctorId() == this.doctorId (Approved)
TC15	Doctor	Doctor.diagnose	Perform diagnosis	Valid Patient, String diagnosis	Prints diagnosis message to console (Approved)
TC16	Doctor	Doctor.updateMedicalHistory	Doctor updates patient history	Clinic, Patient, MedicalHistory	Calls patient.addMedicalRecord & clinic.savePatientChanges (Approved)
TC17	Doctor	Doctor.setSpecialty	Update specialty	String "Neurology"	Specialty updated in object (Failed)
TC18	Doctor	Doctor.createPrescription	Create prescription object	Valid Patient, meds, dosage	Returns Prescription object with correct patientId (Approved)

TC19	Doctor	Clinic.saveDoctorChanges	Persist doctor updates	Modified doctor list	JSON file updated on disk (Approved)
TC20	Doctor	Doctor.toString	String representation	Doctor object	Returns string starting with "Doctor{..." (Failed)

Module 3: Billing Module

ID	Module	Function / Method	Description	Input / Condition	Expected Result
TC21	Billing	Billing.generateBill	Generate initial bill	Billing object instantiated	Status set to "Pending", prints generation msg (Approved)
TC22	Billing	Clinic.saveNewBill	Save bill to repository	Valid Billing object	Bill added to billings list & JSON saved (Approved)
TC23	Billing	Patient.payBill	Patient pays bill	Valid Billing object	Calls bill.updatePayment("Pagado") (Approved)
TC24	Billing	Billing.updatePayment	Update status manually	String "Cancelled"	Status field updates to "Cancelled" (Approved)

TC25	Billing	Clinic.searchBill	Search bill by ID	Valid billId	Returns correct Billing object (Approved)
TC26	Billing	Clinic.searchBill	Search missing bill	Invalid billId	Returns null (Approved)
TC27	Billing	Billing.viewBill	Display bill details	Existing bill	Prints ID, Amount, Status to console (Approved)
TC28	Billing	Clinic.saveBillingChanges	Save modified bills	List of billings	JSON file updated (Approved)
TC29	Billing	Billing.setAmount	Set bill amount	Double value	Amount updated (Approved)
TC30	Billing	Billing.getStatus	Check default status	New Billing (via constructor)	Returns status provided in constructor (Approved)

Module 4: Medical Records

ID	Module	Function / Method	Description	Input / Condition	Expected Result
TC31	History	Doctor.updateMedicalHistory	Add record via Doctor	MedicalHistory object	Record added to Patient's list (Approved)

TC32	History	Patient.viewMedicalHistory	View populated history	Patient with 1+ records	Prints details of all MedicalHistory objects (Approved)
TC33	History	MedicalHistory.toString	Verify output format	MedicalHistory object	Returns string with Allergies, Diseases, Treatments (Approved)
TC34	History	MedicalHistory.setAllergies	Update allergies	String "Penicillin"	Allergies field updated (Approved)
TC35	History	MedicalHistory.setDate	Set record date	Valid Date object	Date field updated (Failed)
TC36	History	Clinic.savePatientChanges	Persist history	Patient with new record	JSON saves the nested medicalHistory list (Approved)
TC37	History	MedicalHistory.addRecord	Log addition	(Method calls sysout)	Prints "Historial ID: [id]" (Approved)
TC38	History	MedicalHistory.getTreatments	Retrieve treatments	Existing record	Returns correct treatment string (Approved)
TC39	History	Patient.getMedicalHistory	Get raw list	Patient object	Returns List<MedicalHistory> (Approved)

TC40	History	MedicalHistory.setObservations	Add observations	String "Patient improving"	Observations field updated (Approved)
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Module 5: Appointment Scheduling

ID	Module	Function / Method	Description	Input / Condition	Expected Result
TC41	Schedule	Receptionist.createAppointment	Create appointment	Clinic, IDs, Date	Returns Appointment, calls clinic.saveNewAppointment (Approved)
TC42	Schedule	Appointment.reschedule	Reschedule appointment	New Date object	Date updated, Status set to "Agendada" (Approved)
TC43	Schedule	Appointment.cancel	Cancel appointment	Existing appointment	Status updates to "Cancelada" (Approved)
TC44	Schedule	Appointment.markCompleted	Complete appointment	Existing appointment	Status updates to "Completeda" (Approved)
TC45	Schedule	Clinic.saveNewAppointment	Save to repo	Valid Appointment	Added to appointments list & JSON saved (Approved)

TC46	Schedule	Appointment.toString	Verify print format	Appointment object	Prints block with ID, Date, PatientID, DoctorID (Approved)
TC47	Schedule	Appointment.setStatus	Manual status override	String "No Show"	Status updated (Approved)
TC48	Schedule	Date.toString	Verify Date format	Date (2025, 10, 20, 14, 30)	Returns "2025-10-20 @ 14:30" (Approved)
TC49	Schedule	Date.setTime	Change time only	Hour 15, Minute 00	Hour/Minute updated, Year/Month/Day remain (Approved)
TC50	Schedule	Receptionist.createAppointment	Verify persistence	After creation	clinic.appointments size increases by 1 (Approved)

Module 6: Appointment Notifications

ID	Module	Function / Method	Description	Input / Condition	Expected Result
TC51	Notification	Notification.send	Send notification	Valid message and date	Prints "ENVIANDO NOTIFICACION " and message (Approved)

TC52	Notificat ion	Notification.setMessage	Update message content	New String content	Message field updated(Approved)
TC53	Notificat ion	Notification.setDate	Set notification date	Valid Date object	Date field updated(Approved)
TC54	Notificat ion	Notification.toString	String representation	Notification Object	Returns string "Notification{...}"(Fai led)
TC55	Notificat ion	Notification	Empty message	Msg "Hello", Date	Object created with correct attributes(Approved)
TC56	Notificat ion	Notification.send	Verify print format	Message	Prints empty message line (Approved)
TC57	Notificat ion	Notification.getMessage	Retrieve content	Existing notification	Returns message string(Approved)
TC58	Notificat ion	Appointment.reschedule	Reschedule output	calling reschedule()	Prints "Cita [ID] reagendada para..." (acting as notification)(Failed)
TC59	Notificat ion	Appointment.cancel	Cancel output	calling cancel()	Prints "Cita [ID] ha sido cancelada.(Approv ed)"

TC60	Notificat ion	requestAppointment	Request log	calling requestAppo intment()	Prints "Paciente [Name] esta solicitando una cita." (Failed)
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Module 7 : Prescriptions Management

ID	Module	Function / Method	Description	Input / Condition	Expected Result
TC61	Prescrip tions	createPrescription	Doctor creates prescription	Valid Patient, Meds	Returns Prescription with default instruction "Tomar cada 8 horas" (Approved)
TC62	Prescrip tions	Clinic.saveNewPrescripti on	Save prescription	Valid Prescription	Added to prescriptions list & JSON saved (Approved)
TC63	Prescrip tions	Prescription.printPrescri ption	Print format	Existing prescription	Prints block with "--- Imprimiendo Prescripcion ---" (Approved)
TC64	Prescrip tions	Prescription.setDo sage	Update dosage	String "500mg"	Dosage updated (Failed)

TC65	Prescriptions	Prescription.setInstructions	Custom instructions	String "Take with food"	Instructions updated (Approved)
TC66	Prescriptions	Prescription.getPatientId	Verify Patient Link	Prescription object	Returns correct patientId (Approved)
TC67	Prescriptions	Clinic.getPrescriptionRepository	Access repository	Clinic instance	Returns non-null JsonRepository (Approved)
TC68	Prescriptions	Prescription.setDate	Set issuance date	Valid Date object	Date updated (Approved)
TC69	Prescriptions	Prescription.setMedication	Update medication name	String "Ibuprofen"	Medication field updated (Approved)
TC70	Prescriptions	Doctor.createPrescription	Verify automatic ID mapping	patient.getPatientId()	Prescription patientId matches patient.getPatientId() (Approved)